

Cross-Sport Video Talent Scout - Product Requirement Document

1. Product Summary

Cross-Sport Video Talent Scout is an AI-powered scouting system that analyzes athlete videos and recommends suitable sports based on movement patterns, posture, coordination, and athletic traits.

2. Problem Statement

Traditional scouting methods rely heavily on physical trials and subjective evaluations, limiting access for rural athletes and overlooking cross-sport talent opportunities.

3. Product Goals

- Analyze smartphone videos using AI
- Predict sport suitability
- Provide explainable recommendations
- Support remote and rural scouting
- Generate text, image, and audio outputs

4. Inputs

- Athlete video upload
- Optional metadata: age, gender, height, weight
- Camera and environment information

5. Outputs

- Performance score
- Top-3 sport recommendations
- Sport-family rankings
- Confidence scores
- Movement trait analysis
- Text and audio explanation

6. Functional Requirements

P0 Requirements:

- Video upload
- Pose estimation
- Feature extraction
- Sport ranking system
- Explainable outputs

P1 Requirements:

- Audio narration
- Coach dashboard
- Batch uploads

P2 Requirements:

- Offline inference
- Multilingual support

7. Machine Learning Pipeline

1. Video quality assessment
2. Pose estimation
3. Temporal motion analysis
4. Trait scoring
5. Sport-fit ranking
6. Explanation generation

8. Privacy and Safety

- Require user consent
- Support data deletion
- Avoid identity recognition
- Use anonymized datasets

9. Final Definition

The product enables fair, scalable, and data-driven sports talent discovery using smartphone videos and AI-powered movement analysis.